SQL Assignment: Beauty Salon Booking Database

Create a simple relational database for a beauty salon booking system. The goal is to practice creating tables, relationships, primary keys, foreign keys, basic constraints, and SQL queries.

1. Database Tables

• **clients**

– client_id

– first_name

– last_name

– email

– phone

• **employees**

– employee_id

– first_name

– last_name

– email

– phone

• **services**

– service_id

– service_name

– description

– duration

– price

• **appointments**

– appointment_id

– client_id

– employee_id

– service_id

– appointment_date

– appointment_time

– status

• **payments**

– payment_id

– appointment_id

– amount

– payment_method

– payment_date

– payment_status

2. Relationships

• A client can book many appointments.

• An employee can have many appointments.

• A service can be included in many appointments.

• Each appointment belongs to one client.

• Each appointment is assigned to one employee.

• Each appointment is for one service.

• Each appointment can have one payment.

• The relationships between tables must be implemented using foreign keys.

3. Required Constraints

• Every table must have a PRIMARY KEY.

• Client email must be UNIQUE and NOT NULL.

• Employee email must be UNIQUE and NOT NULL.

• Service name must be NOT NULL.

• Service price must be greater than or equal to 0.

• Service duration must be greater than 0.

• All foreign keys must reference the appropriate tables.

• Payment amount must be greater than or equal to 0.

• Appointment status must contain a valid value, such as 'Scheduled', 'Completed', or 'Cancelled'.

• Payment status must contain a valid value, such as 'Paid', 'Pending', or 'Refunded'.

4. Sample Data

• 5 clients

• 3 employees

• 5 services

• 8 appointments

• 6 payments

5. Required SQL Queries

• Display all clients.

• Display all available services with their prices.

• Display all appointments with client names, employee names, and service names.

• Display all appointments for a selected client.

• Display all appointments assigned to a selected employee.

- Count how many appointments each employee has.
- Display services with a price greater than 5000.
- Display all appointments ordered by appointment date.
- Display all completed appointments.
- Display all payments with their appointment details.